Do bilinguals avoid ambiguity or minimise processing effort? Insights from an online eye-tracking study in spoken Mandarin

Abstract: Previous research shows that bilinguals tend to choose more explicit referential forms (e.g., overt pronouns over null pronouns) compared to monolingual speakers, but the mechanisms driving this tendency remain debated. By conducting two experiments examining lexical ambiguity in spoken Mandarin, we tested two hypotheses:

Hypothesis 1: bilinguals would rather be redundant than ambiguous in general;

Hypothesis 2: bilinguals avoid ambiguity only when doing so helps reduce cognitive load.

In Experiment 1, 24 L1 Mandarin L2 English speakers in the UK and 23 more-monolingual speakers in China completed a picture naming task (Figure 1), where we manipulated whether the context in which a picture is named makes the preferred label ambiguous (e.g., do speakers avoid saying “fen3 si1” when describing a picture of glass noodles when it appears alongside a picture of fans which shares the same label?). All speakers were tested in their native language. Experiment 2 extended this task by incorporating online eye-tracking using WebGazer. Forty-four L1 Mandarin L2 English bilinguals and 41 more-monolinguals were included in the verbal response analysis. After checking eye-tracking calibration quality in two tasks across two sessions, 29 bilinguals and 36 more-monolinguals were retained for the eye-movement (i.e. saccade) analysis. We analysed data using Bayesian mixed-effects logistic regression models.

We combined participants’ verbal responses across two experiments. Our results showed that, contrary to Hypothesis 1, bilinguals were more likely than more-monolinguals to use ambiguous expressions for ambiguous trials (Figure 2). Eye-tracking analysis further revealed that bilinguals tended to direct early attention toward image pairs with more accessible labels, indicating a preference for linguistic choices that are cognitively less demanding; by contrast, more-monolinguals showed proactive monitoring of ambiguity depending on their responses (Figure 3). These findings support Hypothesis 2 and shed light on cognitive constraints in bilingual lexical access and early signs of lexical attrition. One possible explanation is that the presence of a competitor image that shares linguistic features with the target likely activates a shared label, making it more accessible. To avoid such ambiguity, speakers need to inhibit this activated label and instead retrieve a less active, more specific alternative. This retrieval process becomes more effortful for bilinguals due to their lexical access difficulties. As such, it is more cognitively efficient to rely on the shared, ambiguous label that is already activated in the mental lexicon, especially when there is no immediate pressure from a conversational partner.

Figure 1: An example of the Segment-Only Overlap ambiguity category: target is *hua1 ban4* (*花瓣*), petals, competitor is *hua2 ban3 (滑板),* skateboard, filler is *qing1 ting2 (蜻蜓),* dragonfly. In the unambiguous condition, the competitor is replaced by a new filler.

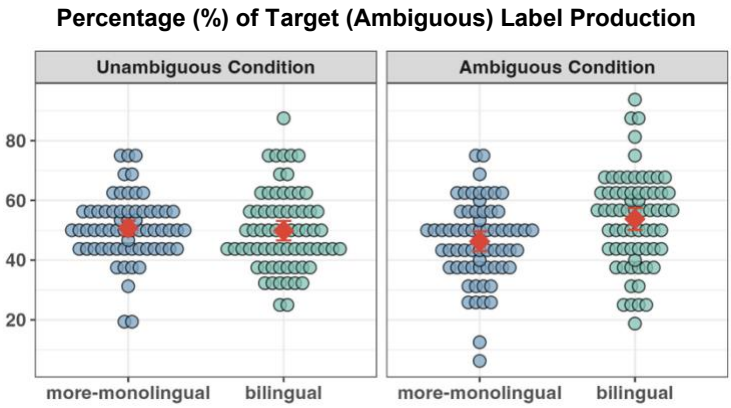

Figure 2: Target label production across conditions and groups, with data combined across experiments. Target label refers to descriptions that match the default labels. Each dot represents the % of target labels produced by one speaker; the red diamond indicates the mean of by-participant percentage, with error bars showing bootstrapped 95% confidence intervals of the mean.

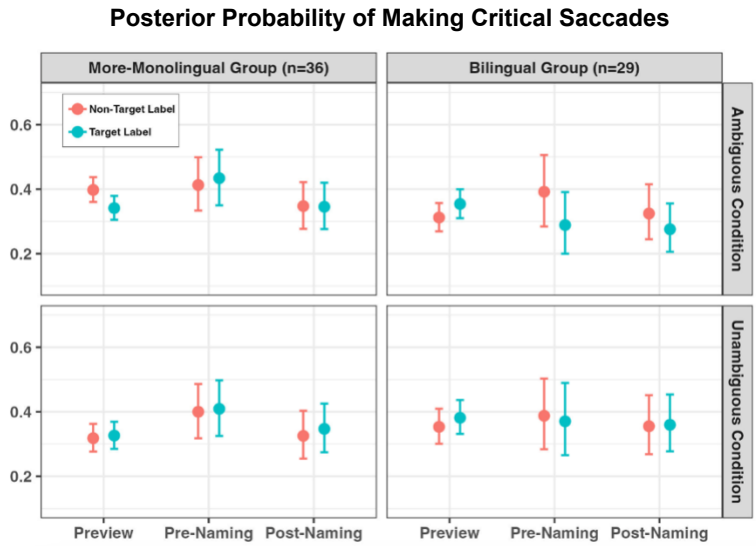

Figure 3: The conditional effects of Phase, Condition, Group, and Response Type (target vs non-target) on the posterior probability of critical saccades (Ambiguous Condition: target-competitor; Unambiguous Condition: target-new filler), as estimated by the Bayesian model. The red and blue dots represented estimated mean probability, and error bars indicate the 95% credible intervals.